

Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) Checklist

SECTION	ITEM	PRISMA-ScR CHECKLIST ITEM	REPORTED ON PAGE #
TITLE			
Title	1	Identify the report as a scoping review.	The scoping review is reported as Section 2.1 ("Existing literature: Scoping review") within this larger validation article; the article as a whole is not titled as a scoping review, as it also reports three empirical validation studies.
ABSTRACT			
Structured summary	2	Provide a structured summary that includes (as applicable): background, objectives, eligibility criteria, sources of evidence, charting methods, results, and conclusions that relate to the review questions and objectives.	The abstract briefly references the scoping review ("Based on a comprehensive scoping review of existing ethics measurement instruments...") but does not provide a full structured summary specific to the scoping review; eligibility criteria, sources of evidence, and charting methods are not separately abstracted.
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known. Explain why the review questions/objectives lend themselves to a scoping review approach.	Section 2.1, opening paragraph: "To the best of our knowledge, no quantitative, multidimensional instrument currently exists that integrates multiple ethical theories and is specifically designed for the ethical assessment of technologies," explaining the rationale relative to the Multidimensional Ethics Scale (MES) and prior literature.
Objectives	4	Provide an explicit statement of the questions and objectives being addressed with reference to their key elements (e.g., population or participants, concepts, and context) or other relevant key elements used to conceptualize the review questions and/or objectives.	Section 2.1: "The purpose of our scoping review was therefore threefold: (a) to identify existing ethics scales and their theoretical underpinnings, (b) to extract and synthesize central publications discussing the limitations and extensions of existing scales, and (c) to assess whether ethics scales have previously been applied within the context of technology assessment."
METHODS			
Protocol and registration	5	Indicate whether a review protocol exists; state if and where it can be accessed (e.g., a Web address); and if available, provide registration information, including the registration number.	No study-specific protocol was registered (e.g., PROSPERO or OSF). The review followed the JBI methodology guidelines for scoping reviews (Tricco et al., 2018), as stated in Section 2.1.
Eligibility criteria	6	Specify characteristics of the sources of evidence used as eligibility criteria (e.g., years considered, language, and publication status), and provide a rationale.	Section 2.1, "Eligibility criteria," and Table 1 (inclusion/exclusion criteria): citation-anchored to Reidenbach and Robin (1990), English-language sources, 1990 through March 2022, journal articles plus grey literature (e.g., conference abstracts, blogs); rationale

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			given for the citation-anchored, MES-focused scope.
Information sources*	7	Describe all information sources in the search (e.g., databases with dates of coverage and contact with authors to identify additional sources), as well as the date the most recent search was executed.	Section 2.1, "Search strategy": Google Scholar and Web of Science. Coverage extends through March 2022 (date of the most recent included article); the exact date the search was executed is not separately reported.
Search	8	Present the full electronic search strategy for at least 1 database, including any limits used, such that it could be repeated.	Section 2.1, "Search strategy": citation search for Reidenbach and Robin (1990) in Google Scholar and Web of Science; Google Scholar records were limited to those containing ("review" OR "meta analysis") AND "model"; supplemented by backward reference searching. The full literature database is available in the "Scoping Review/database raw data" folder of the public repository.
Selection of sources of evidence†	9	State the process for selecting sources of evidence (i.e., screening and eligibility) included in the scoping review.	Section 2.1, "Study selection process": a three-phase process (identification and deduplication; title/abstract/partial full-text screening; full-text-based inclusion into four categories), with the first two phases conducted in Rayyan (Ouzzani et al., 2016).
Data charting process‡	10	Describe the methods of charting data from the included sources of evidence (e.g., calibrated forms or forms that have been tested by the team before their use, and whether data charting was done independently or in duplicate) and any processes for obtaining and confirming data from investigators.	Section 2.1, "Data extraction and data synthesis": year of publication, central arguments, relation to the Multidimensional Ethics Scale, and publication type were extracted per article (Appendix, "Sources of evidence" table).
Data items	11	List and define all variables for which data were sought and any assumptions and simplifications made.	Year of publication, central arguments for the development of ESTA, relation to the "Multidimensional Ethics Scale," and publication type (Section 2.1; Appendix, "Sources of evidence" table).
Critical appraisal of individual sources of evidence§	12	If done, provide a rationale for conducting a critical appraisal of included sources of evidence; describe the methods used and how this information was used in any data synthesis (if appropriate).	Not conducted. A formal critical appraisal (e.g., risk-of-bias assessment) of included sources was not performed, consistent with PRISMA-ScR guidance that this is optional for scoping reviews given their descriptive-mapping purpose.
Synthesis of results	13	Describe the methods of handling and summarizing the data that were charted.	Section 2.1, "Findings of included studies" (Results): narrative synthesis of the extracted data, organized into four categories (Overview, Multidimensional Ethics Scale, Further Scales,

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			Application) and their sub-categories; full categorization in the Appendix "Sources of evidence" table.
RESULTS			
Selection of sources of evidence	14	Give numbers of sources of evidence screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally using a flow diagram.	Section 2.1, "Search findings": 906 records retrieved from the two databases; 325 duplicates removed; titles and abstracts screened; 80 articles full-text screened (76 database-identified plus 4 via backward reference searching); 40 studies included. Flow diagram: PRISMA-ScR flow chart figure in Section 2.1.
Characteristics of sources of evidence	15	For each source of evidence, present characteristics for which data were charted and provide the citations.	Appendix, "Sources of evidence" table: category, subcategory, citation, and central arguments/relation to the MES for each of the 40 included studies.
Critical appraisal within sources of evidence	16	If done, present data on critical appraisal of included sources of evidence (see item 12).	Not applicable — no critical appraisal was conducted (see item 12).
Results of individual sources of evidence	17	For each included source of evidence, present the relevant data that were charted that relate to the review questions and objectives.	Section 2.1, "Findings of included studies," and Appendix "Sources of evidence" table present the extracted data (category, central arguments, MES relation) for each included source.
Synthesis of results	18	Summarize and/or present the charting results as they relate to the review questions and objectives.	Section 2.1, "Findings of included studies": narrative synthesis organized by the four categories/sub-categories above, focused on studies relevant to ESTA's construction; not all identified ethics scales are elaborated on in the main text (full list in the Appendix).
DISCUSSION			
Summary of evidence	19	Summarize the main results (including an overview of concepts, themes, and types of evidence available), link to the review questions and objectives, and consider the relevance to key groups.	The scoping review's findings are synthesized in Section 2.1 and used directly to motivate ESTA's theoretical scope in Section 2.2 ("These considerations resulted in an ethics scale that covers deontology, utilitarianism, hedonism, contractualism, and virtue ethics... as well as relativism"). The article does not include a separate Discussion subsection for the scoping review alone, as it is one component of a larger validation study.
Limitations	20	Discuss the limitations of the scoping review process.	Section 2.1, "Eligibility criteria," explicitly discusses the scope limitation of the citation-anchored search strategy: the review "should be interpreted as a focused mapping of the MES-related psychometric literature rather than an exhaustive review of the broader technology-ethics literature." Appendix

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			(extended literature review) further situates this scope against adjacent AI-ethics, trust, and socio-technical literatures.
Conclusions	21	Provide a general interpretation of the results with respect to the review questions and objectives, as well as potential implications and/or next steps.	The scoping review's conclusion is integrated into Section 2.2: it directly motivated the development of a new scale, rather than reuse of the MES, covering six ethical theories and addressing the substantive and statistical critiques of the MES identified in the review.
FUNDING			
Funding	22	Describe sources of funding for the included sources of evidence, as well as sources of funding for the scoping review. Describe the role of the funders of the scoping review.	Deutsche Forschungsgemeinschaft (DFG) under Germany's Excellence Strategy (EXC 2193/1 – 390951807) and the Eva Mayr-Stihl Foundation.

JB1 = Joanna Briggs Institute; PRISMA-ScR = Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews.

* Where *sources of evidence* (see second footnote) are compiled from, such as bibliographic databases, social media platforms, and Web sites.

† A more inclusive/heterogeneous term used to account for the different types of evidence or data sources (e.g., quantitative and/or qualitative research, expert opinion, and policy documents) that may be eligible in a scoping review as opposed to only studies. This is not to be confused with *information sources* (see first footnote).

‡ The frameworks by Arksey and O'Malley (6) and Levac and colleagues (7) and the JBI guidance (4, 5) refer to the process of data extraction in a scoping review as data charting.

§ The process of systematically examining research evidence to assess its validity, results, and relevance before using it to inform a decision. This term is used for items 12 and 19 instead of "risk of bias" (which is more applicable to systematic reviews of interventions) to include and acknowledge the various sources of evidence that may be used in a scoping review (e.g., quantitative and/or qualitative research, expert opinion, and policy document).

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